AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational Databases

Midterm Examination (MTA)
Spring Semester 2025/2026
Date: 21/ Apr. / 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: 30
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choice questions [5 Marks]

This part consists of 5 questions carrying a weight of 1 mark for each.

1. Which of the following is not a characteristic of the database approach compared to traditional file processing?
   a. Program-data independence
   b. Multi-user support
   c. Data redundancy increase
   d. Support for multiple views

2. In relational algebra, which of the following operations is used to extract specific columns from a relation?
   a. σ (SELECT)
   b. ⋈ (JOIN)
   c. π (PROJECT)
   d. ∪ (UNION)

3. Given the schema:
   EMPLOYEE(EmpID, Name, ManagerID)
   Which of the following extract of the relational algebra queries retrieves the names of employees who do not have a manager?
   a. σ ManagerID ≠ NULL (EMPLOYEE)
   b. σ ManagerID = NULL (EMPLOYEE)
   c. π Name (EMPLOYEE ⋈ EMPLOYEE.EmpID = ManagerID)
   d. π Name (σ ManagerID = EmpID (EMPLOYEE))

4. In an ER model, a weak entity must always:
   a. Have its own primary key
   b. Be linked to a strong entity via a total and identifying relationship
   c. Be multivalued
   d. Represent a superclass in generalization

5. Consider the relations:
   PRODUCT(ProductID, Name)
   ORDER(OrderID, ProductID)
   Which of the following relational algebra expressions finds ProductIDs that were never ordered?
   a. π ProductID (PRODUCT) - π ProductID (ORDER)
   b. π ProductID (ORDER) - π ProductID (PRODUCT)
   c. π ProductID (PRODUCT ⋈ ORDER)
   d. π ProductID (PRODUCT) ∩ π ProductID (ORDER)

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Part 2: Problem solving questions [25 Marks]

Question 2 [8 Marks]

Consider the following relational schema for a library system. Use SQL to answer the following questions.

MEMBER(MemberID, Name, Age, Gender)

BOOK(BookID, Title, Author, Genre)

LOAN(MemberID, BookID, LoanDate, ReturnDate)

1. Retrieve the names of all members. **[1 mark]**
2. Retrieve the titles and authors of all books in the genre 'History'. **[1 mark]**
3. List the names and ages of all female members. **[1 marks]**
4. Retrieve the titles of books borrowed by 'Alice'. **[1 marks]**
5. Retrieve the distinct names of members who borrowed books written by 'J.K. Rowling'. **[1 marks]**
6. Retrieve the names of members who never borrowed any book. **[3 marks]**

Question 3 [8 Marks]

A national healthcare system maintains records of Healthcare Providers, Individuals, and Medical Services. The system has the following structure:

- Each Healthcare Provider has a unique ProviderID, Name, and Type. A provider is either a Hospital or a Clinic, but not both (partition constraint).
- A Hospital has additional attributes: NumberOfBeds, Accredited (yes/no).
- A Clinic has: Specialty and WalkInOnly (yes/no).
- Individuals are uniquely identified by SSN, and have Name, BirthDate, and IsInsured.
- Each Individual can Receive medical services, and each Service has:
  - ServiceID, ServiceType, Date, and is performed by one Provider.
- A Functional Integrity Constraint (FIC) applies:
  - A provider can offer only one type of medical service.
  - An Inclusion Constraint applies:
  - If an individual receives a service, they must be insured.

1. Draw the ER Diagram representing this scenario. **[5 marks]**
2. Mention three constraints. **[3 marks]**
   a. Partition Constraint
   b. Functional Integrity Constraint (FIC)
   c. Inclusion Constraint

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Question 4 [9 Marks]

A university wants to create a database to manage academic advising sessions between students and faculty.

- Each Advisor has a unique AdvisorID, Name, and Department.
- Each Student has a unique StudentID, Name, and Major.
- A Student can have multiple advising sessions, but each session is conducted by only one Advisor.
- An AdvisingSession has a unique SessionID, Date, Topic, and is associated with one Student and one Advisor.

1. Identify the entities, their attributes, and primary keys. **[3 marks]**
2. Draw the ER diagram representing the above scenario. **[3 marks]**
3. Convert the ER diagram into relational tables (schemas). **[3 marks]**

End of questions

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